

Samet Sağ

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Mechanical and Materials Engineering graduate from Queen's University, now working in process and quality at Toromont CAT. Experience designing tooling, investigating component failures, and improving shop-floor processes.

EDUCATION

Queen's University

Sep. 2020 – May 2025

BASc, Mechanical and Materials Engineering

Kingston, ON

- GPA: 3.5; Dean's Scholar distinction (twice); Student Leader in Experiential Learning Certificate.

EXPERIENCE

Process & Quality Specialist

Feb. 2026 – Present

Toromont CAT

Bradford, ON

- Investigate failed components and quality issues in heavy-equipment remanufacturing. Identify root causes and put corrective actions in place.
- Design tooling and fixtures for production. Write and update SOPs, work instructions, and quality records to meet ISO 9001 requirements.

SparQ Studios Makerspace Supervisor

Feb. 2025 – May 2025

Dunin-Deshpande Queen's Innovation Centre

Kingston, ON

- Helped students use 3D printers, scanners, woodworking tools, and metalworking equipment. Troubleshoot equipment and helped keep the space safe and organized.

Mechanical Engineering Intern

May 2023 – Aug. 2024

Toromont CAT

Concord, ON

- Designed stands, fixtures, and components in Inventor. When drawings were unavailable, measured the parts and reverse-engineered their interfaces.
- Designed and implemented an oil dispensing station that cut daily dispensing labour time by 80%. Managed up to 10 engineering and process projects at a time.
- Wrote 10 SOPs and updated more than 40 documents covering manufacturing processes.

Chassis/Ergonomics & Brake System Team Member

2021–2023; 2024–2025

Queen's Baja SAE

Kingston, ON

- Led brake design for the 2025 vehicle and mentored two junior members. Earlier chassis work included designing and fabricating a detachable bumper, fuel tank mount, and splash guard.
- Used SolidWorks for design and FEA, and Mastercam/CAM for fabrication.

SELECTED PROJECTS

Rehabilitation Robot End-Effectors | *Capstone*

- Designed and prototyped cable-driven end-effectors and a modular attachment for a rehabilitation robot. Checked and refined the geometry using FEA in Fusion 360.

Trailer Hitch Optimization | *ANSYS APDL*

- Adjusted about 25 design variables in an FEA model, reducing maximum stress by 22% and volume by 5%.

TECHNICAL SKILLS & LEADERSHIP

Design & Manufacturing: Inventor, SolidWorks, Fusion 360, ANSYS, Mastercam, GD&T, BOMs, rapid prototyping

Quality & Programming: Root cause analysis, failure analysis, continuous improvement, Python, MATLAB

Competition: Represented Queen's University in consulting at the Ontario Engineering Competition.